

Introduction and Use of TensorFlow

Popular Industry Tool for Machine
Learning & Deep Learning

What is TensorFlow?

- TensorFlow is an open-source platform developed by Google Brain for machine learning (ML) and deep learning (DL).
- Provides a flexible ecosystem of tools, libraries, and community support.
- Designed for both research prototyping and production deployment.

Key Features of TensorFlow

- • Open-source and scalable.
- • Runs on CPUs, GPUs, and TPUs.
- • Supports multi-language APIs (Python, C++, JavaScript).
- • High-level API with Keras integration.
- • Tools for visualization (TensorBoard) and deployment (TFX).

TensorFlow Ecosystem

- TensorFlow Core – Low-level API for flexibility.
- tf.keras – High-level API for quick prototyping.
- TensorFlow Lite – Deployment on mobile and IoT.
- TensorFlow.js – Run models in browsers.
- TensorBoard – Visualize metrics and performance.

Use of TensorFlow in Industry

- Healthcare – Disease prediction, medical imaging.
- Finance – Fraud detection, stock prediction.
- Retail – Recommendation engines.
- Manufacturing – Predictive maintenance.
- Transportation – Autonomous driving and route optimization.

Basic TensorFlow Code Example

- `import tensorflow as tf`
- `x = tf.constant([1.0, 2.0, 3.0])`
- `w = tf.Variable(0.5)`
- `b = tf.Variable(1.0)`
- `y = w * x + b`
- `print('Predicted output:', y.numpy())`

Summary

- TensorFlow is a powerful and flexible ML/DL framework.
- Widely used in industries like healthcare, finance, retail, and transportation.
- Ecosystem includes tools like TensorBoard, TFX, Lite, and JS.
- Keras integration makes it beginner-friendly and production-ready.

